

Reminders

Upcoming due dates

Fri QUIZ?????? Dunno probably not
Discussion lab 5

Next Wed Data checkpoint

Machine Learning I

Data Science in Practice

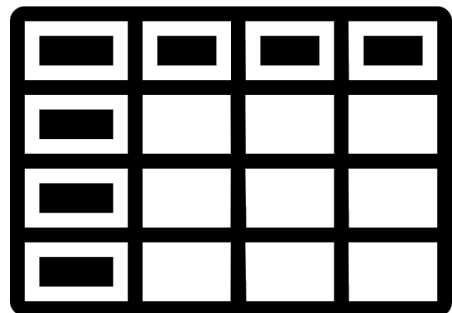
- **Problem:** Detecting whether credit card charges are fraudulent.
- **Data science question:** Can we use the time of the charge, the location of the charge, and the price of the charge to predict whether that charge is fraudulent or not?
- **Type of analysis:** Predictive analysis



Robert Hecht-Neilsen and Zeus (and others) sold HNC for \$810M in 2002
Around here lots of people see him as a major contributor to the
development of neural networks and data science as we know them today

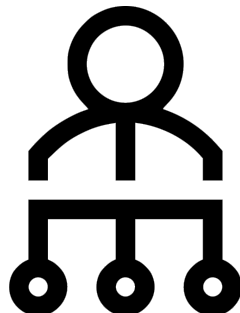
predictive analysis
uses data you have now
to make predictions in
the future

machine learning
approaches are used for
predictive analysis!



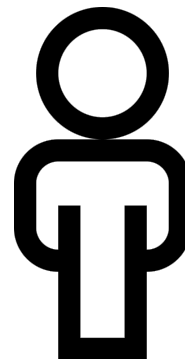
data

train →

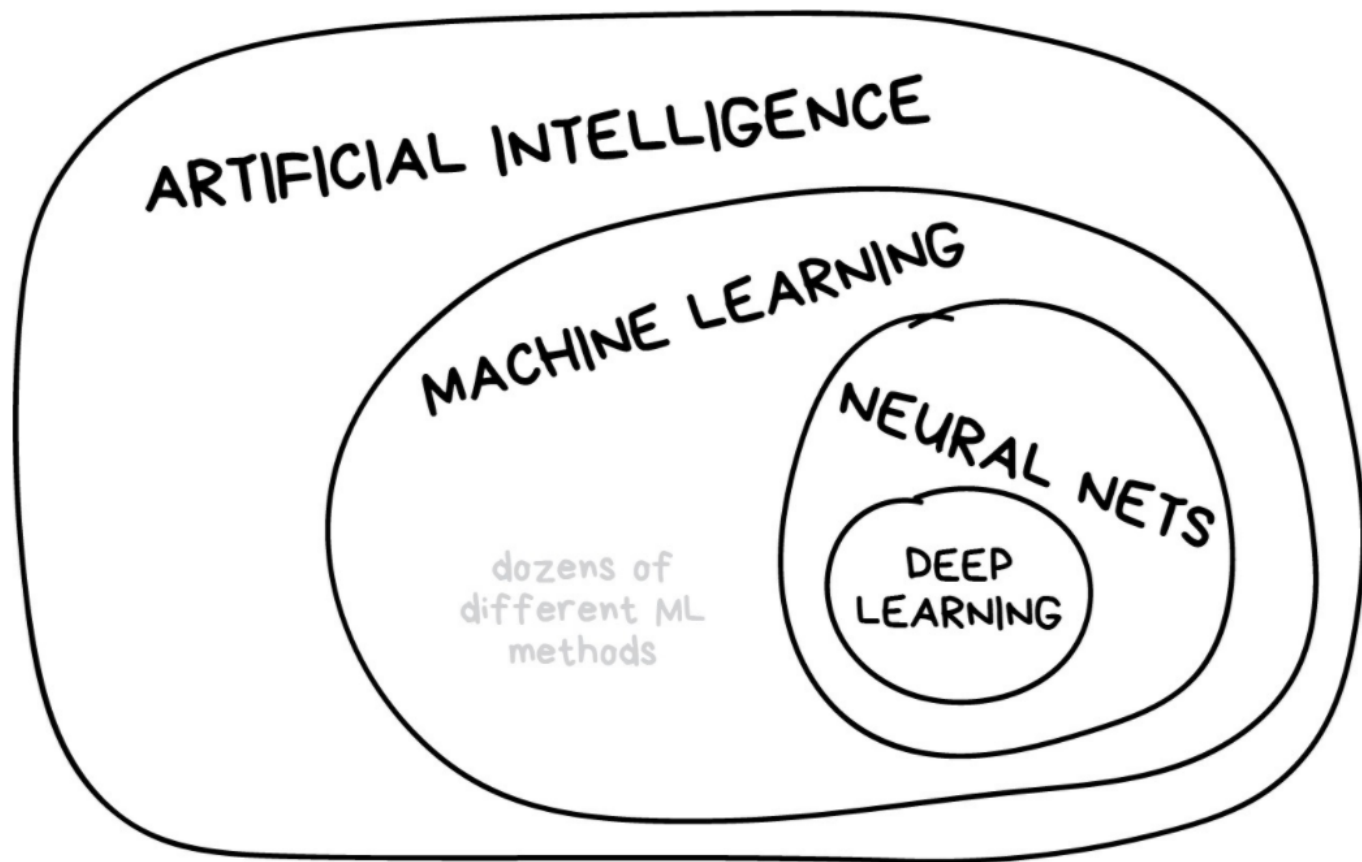


model

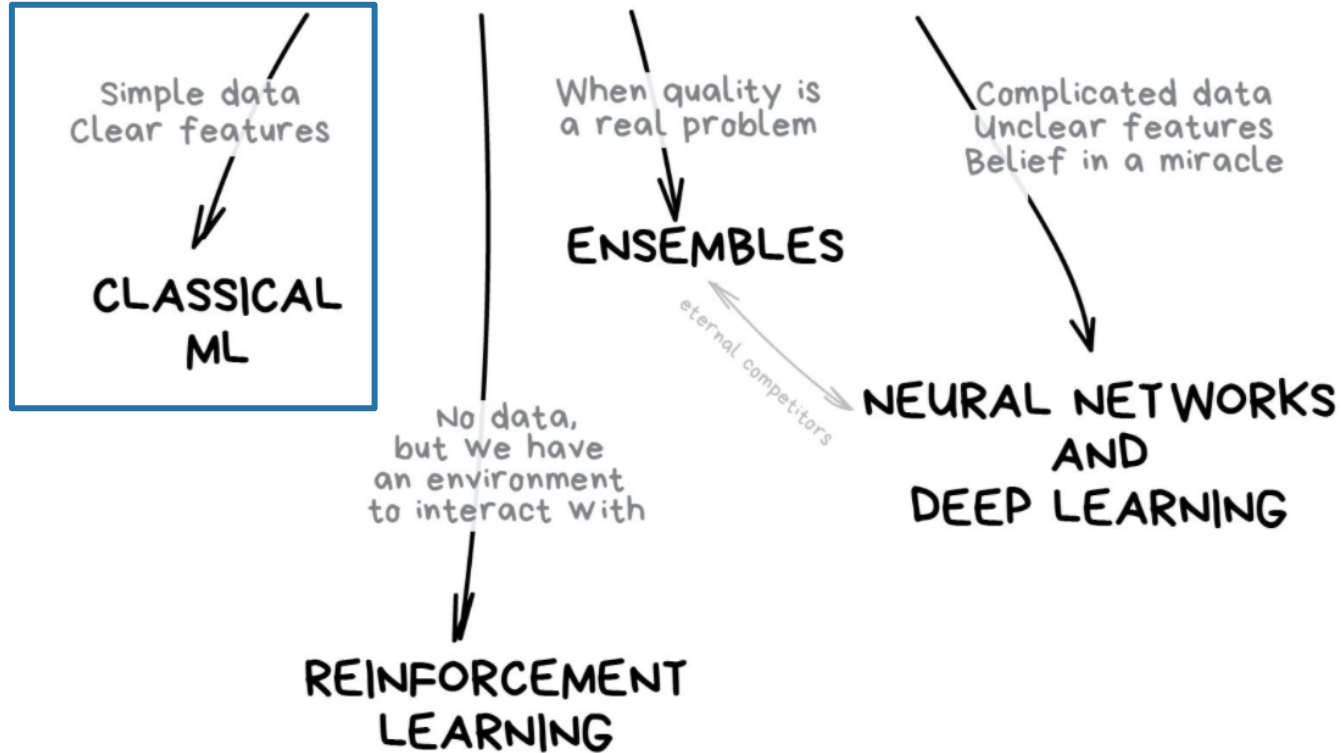
predict →



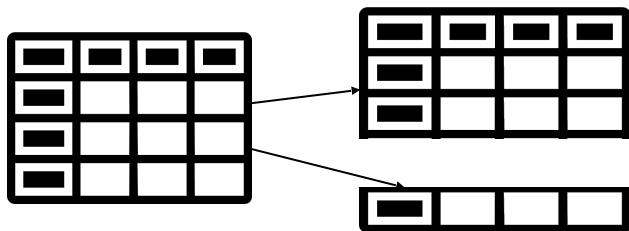
In contrast to statistical approaches which care more about the model accurately reflecting the process than nailing the predictions



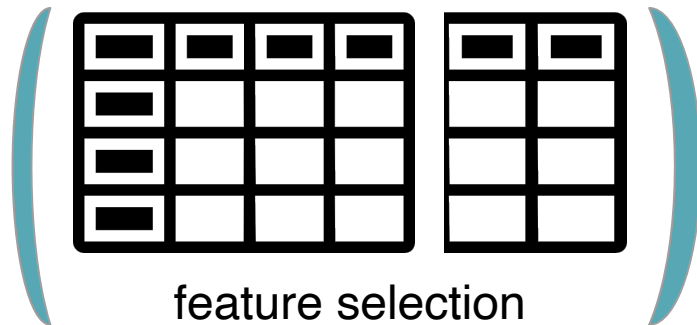
THE MAIN TYPES OF MACHINE LEARNING



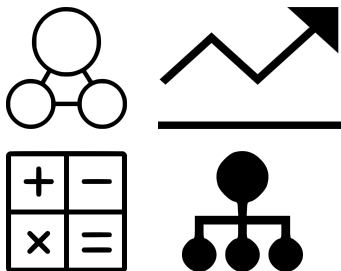
The usual ingredients in the ML recipe



data
partitioning



feature selection



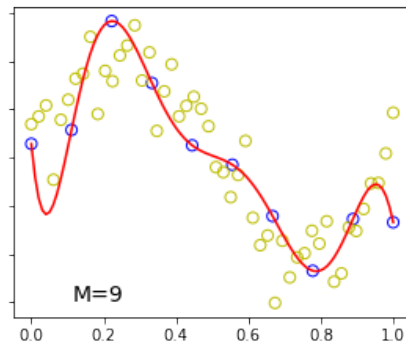
model selection



model assessment

Preventing overfitting

- ML / AI algorithms are often VERY COMPLEX models
 - think of the 9th order model example where 10 data points are fit perfectly with 0 error and are therefore overfit on the noise as well as the signal
 - Now realize that GPT-3.5 has around 175 billion parameters
- Most of ML is preventing powerful models from overfitting when applied to your data
 - Proper data partitioning
 - Regularization in algorithms
 - Data augmentation

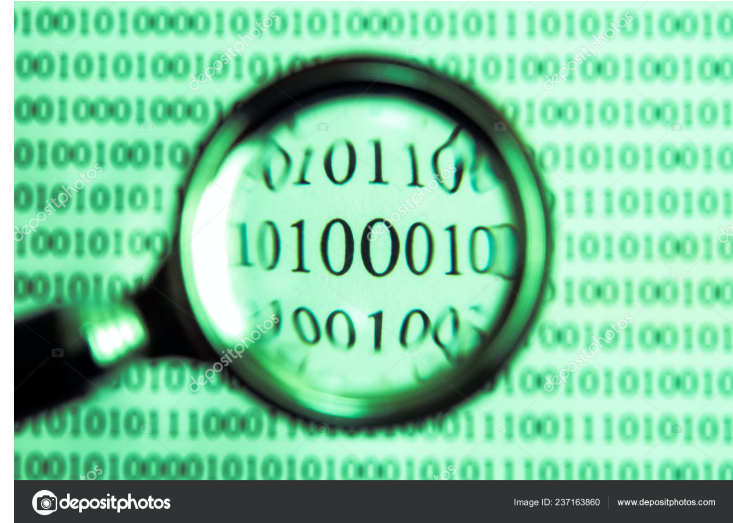


Drowning in data?



Image borrowed from
<https://medium.com/@tmaoconnor/marketing-is-drowning-in-data-yet-its-starving-for-answers-f15ee890a732>

Limited data?



depositphotos

Image ID: 237163860 www.depositphotos.com

Data partitioning:
Evaluating how well you will
generalize to new data with huge
amounts of data

All the data

■	■	■	■
■			
■			
■			

training set

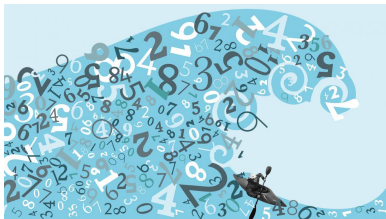
■			
■			

Data used to build
your model

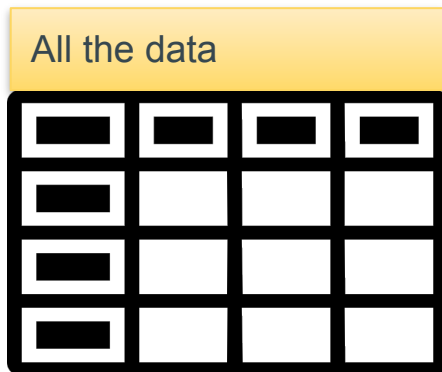
test set

■			
■			

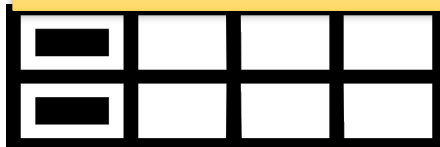
Used to assess how good
the final model is
(Generalization)



Data partitioning:
Finding the best version of your
model through validation



training set



Data used to build
your model

validation set



Fine-tune the model to
increase prediction
accuracy
(Hyper-parameter tuning,
feature selection, model
selection)

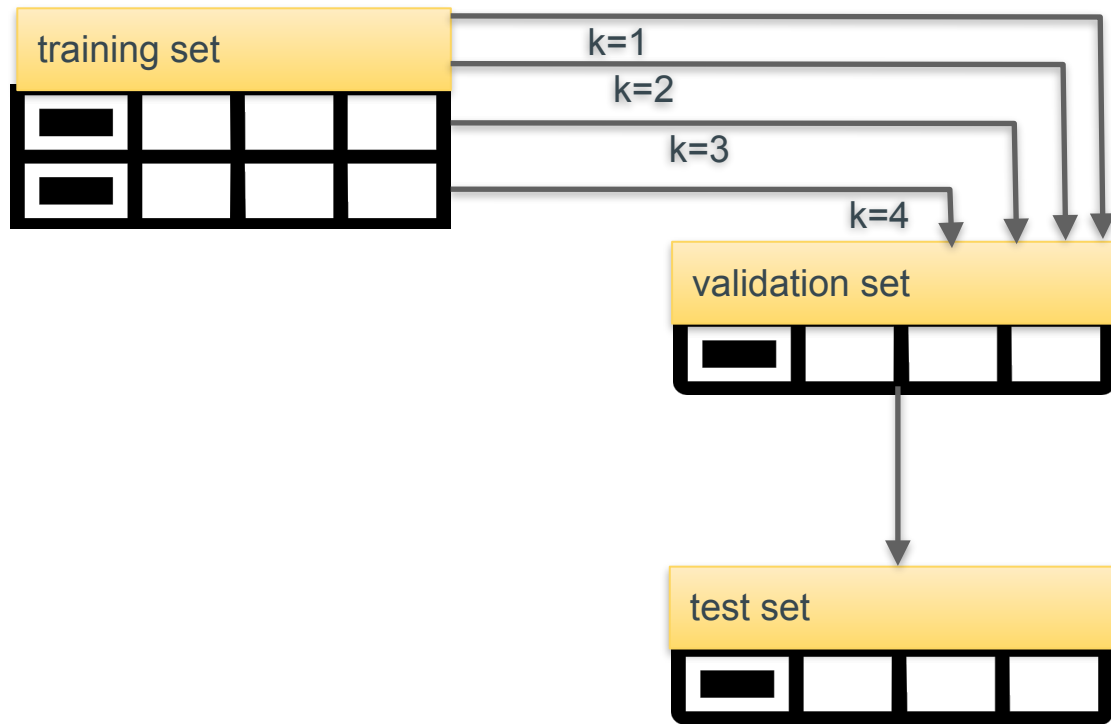
test set



Used to assess how good
the final model is
(Generalization)

Selecting a model

Imagine we are using a ML model with a hyper parameter k taking values from 1 to 4

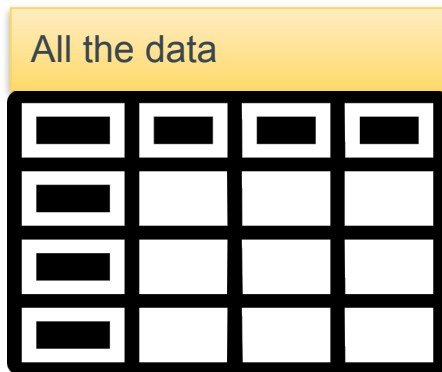


Pick the model that performs best on validation data...

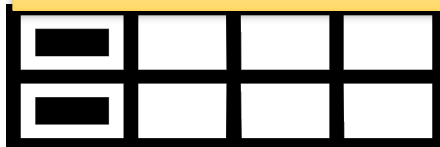
Now retrain that k -value on a combo of training set + validation set

And evaluate its future performance level on the test set

Data partitioning:
Finding the best version of your
model through validation



training set



Typically biggest
chunk
50 - 80%

validation set



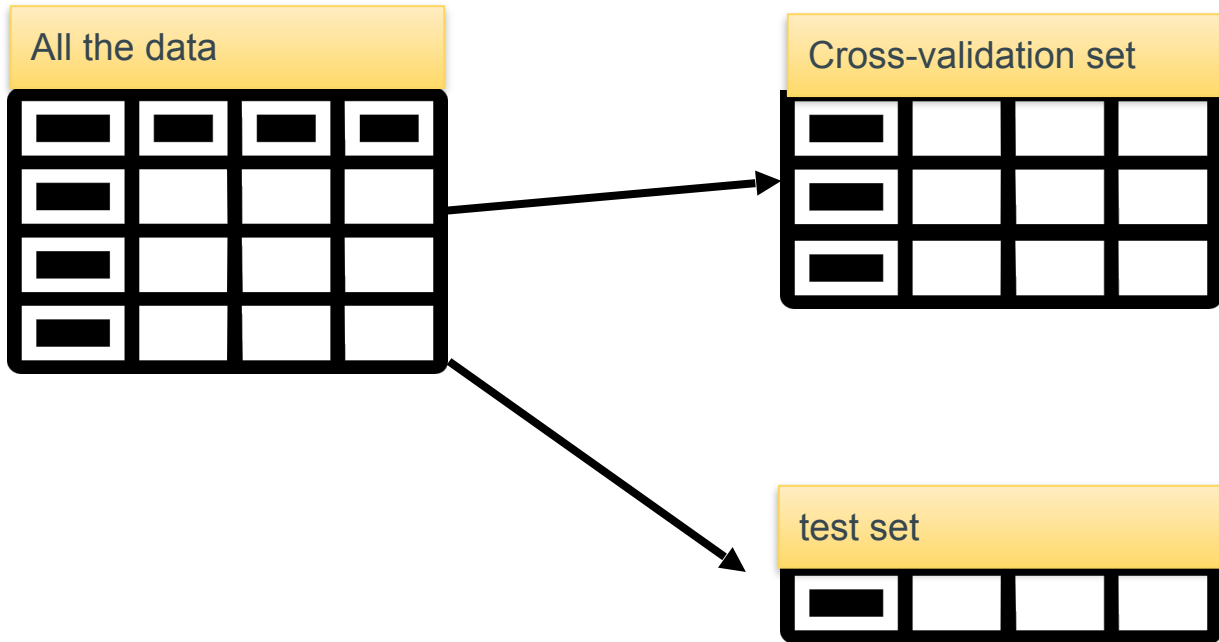
Smaller
10 - 25%

test set



Smaller
10 - 25%

Cross-validation: Being more efficient using small amounts of data



When you don't have huge data

K-fold cross validation



1st Split

	Cross validation set			
Fold 1				
Fold 2				
Fold 3...				
Fold k				

use 1st fold of the data as
validation, train on
everything else

Training set

Validation set

When you don't have huge data

K-fold cross validation



2nd Split

	Cross validation set			
Fold 1				
Fold 2				
Fold 3...				
Fold k				

**use 2nd fold of the data as
validation, train on
everything else**

Training set

Validation set

When you don't have huge data

K-fold cross validation



k-th Split

	Cross validation set			
Fold 1				
Fold 2				
Fold 3...				
Fold k				

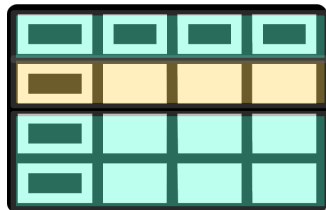
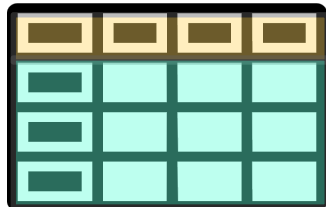
use k-th fold of the data as
validation, train on
everything else

Training set

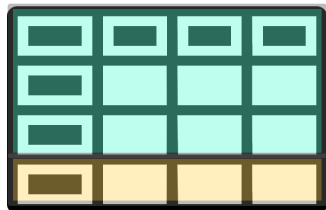
Validation set

When you don't have huge data

K-fold cross validation

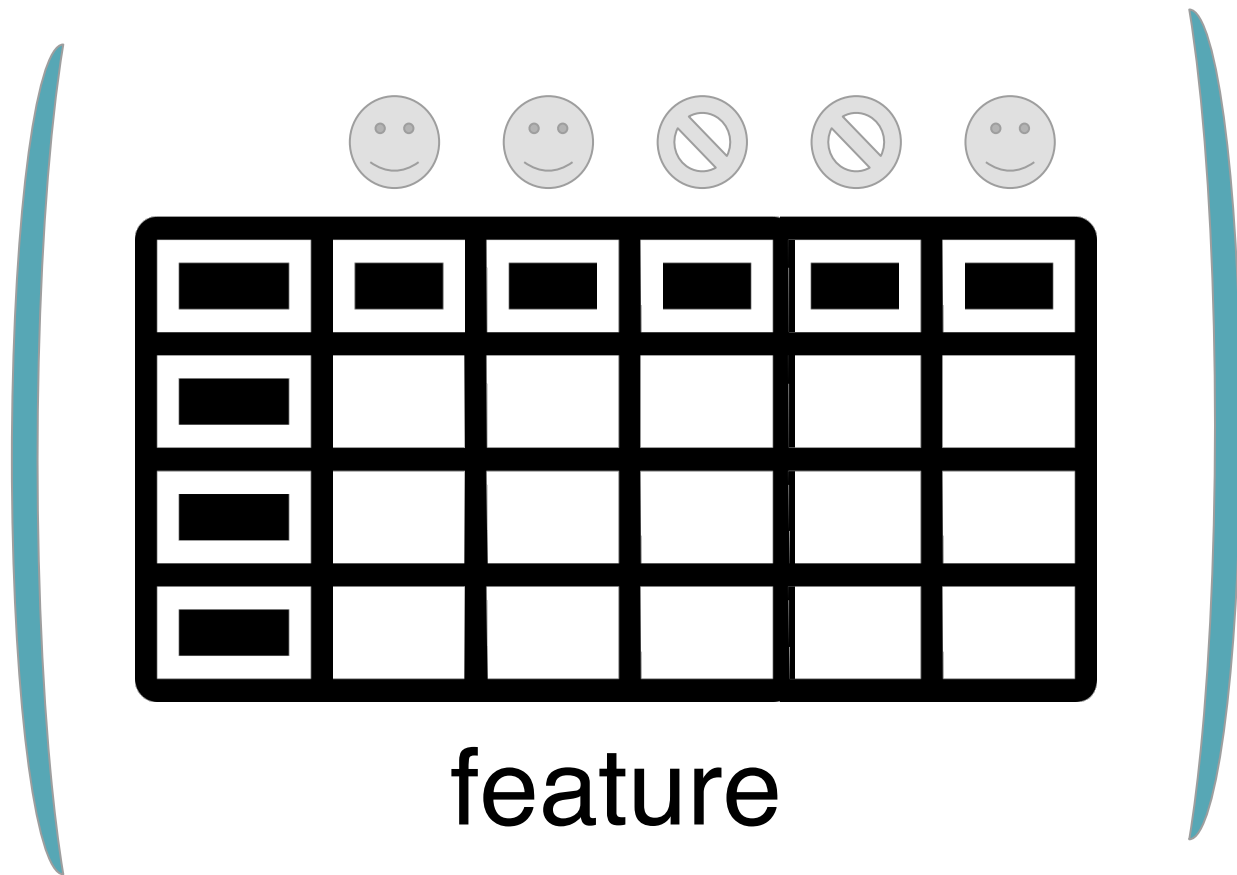


⋮



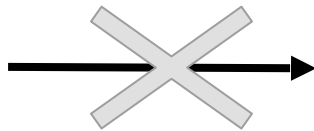
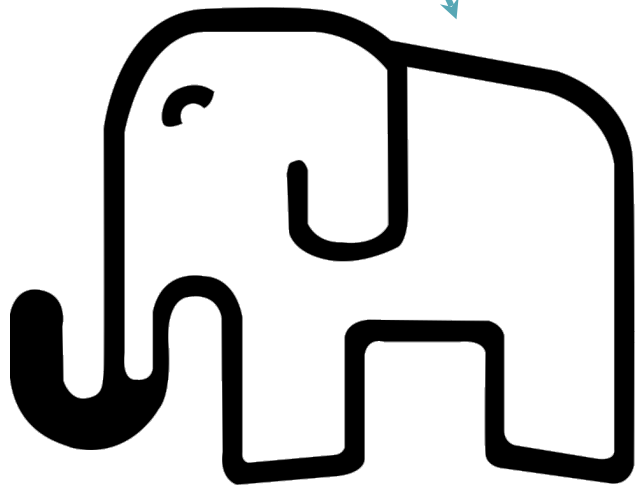
Use the mean of the hold out set performances to estimate either test or validation error

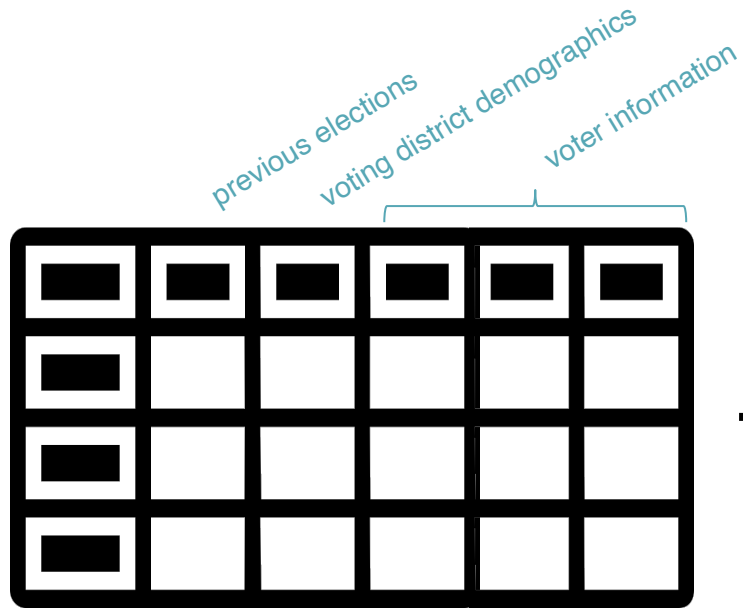




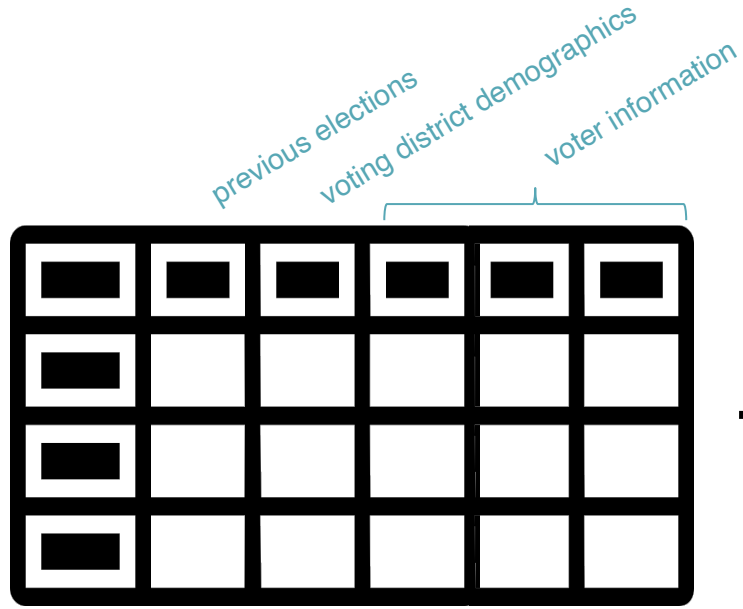
feature
selection

elephant height data
are likely not predictive
of US elections

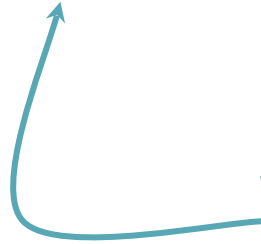




these data are likely
predictive of US
election outcomes



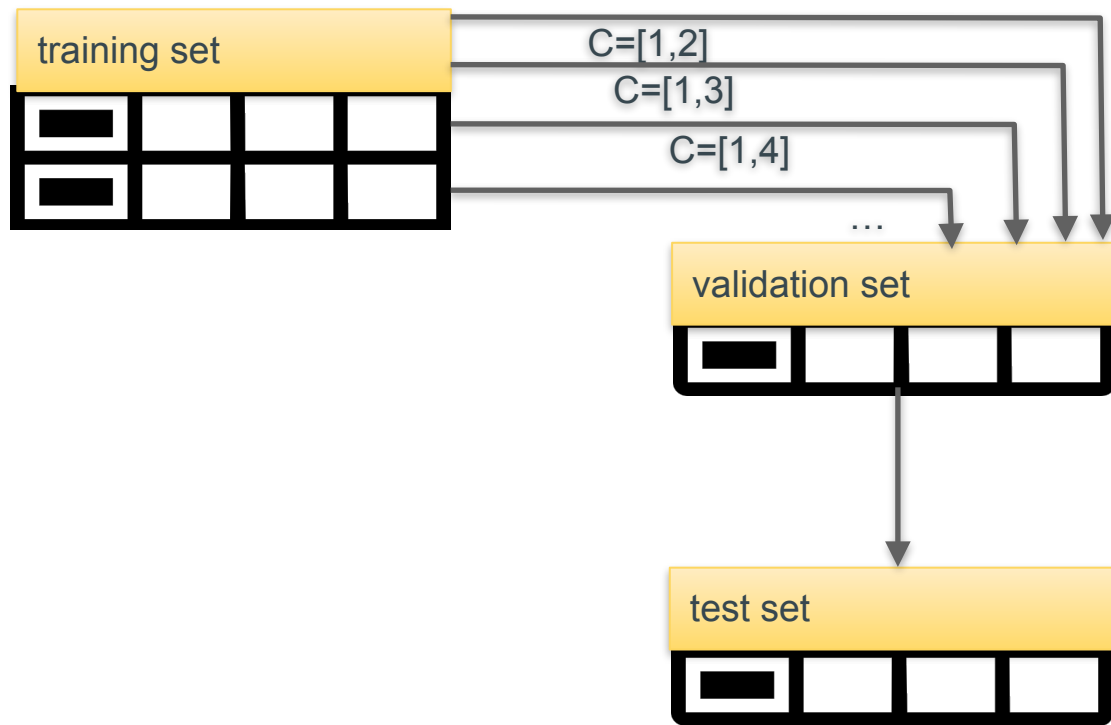
feature selection determines which variables are most predictive and includes them in the model



variables that can be used for accurate prediction exploit the relationship between the variables but do NOT mean that one causes the other

Selecting a model

Feature selecting an ML model with different subsets of the data columns, C



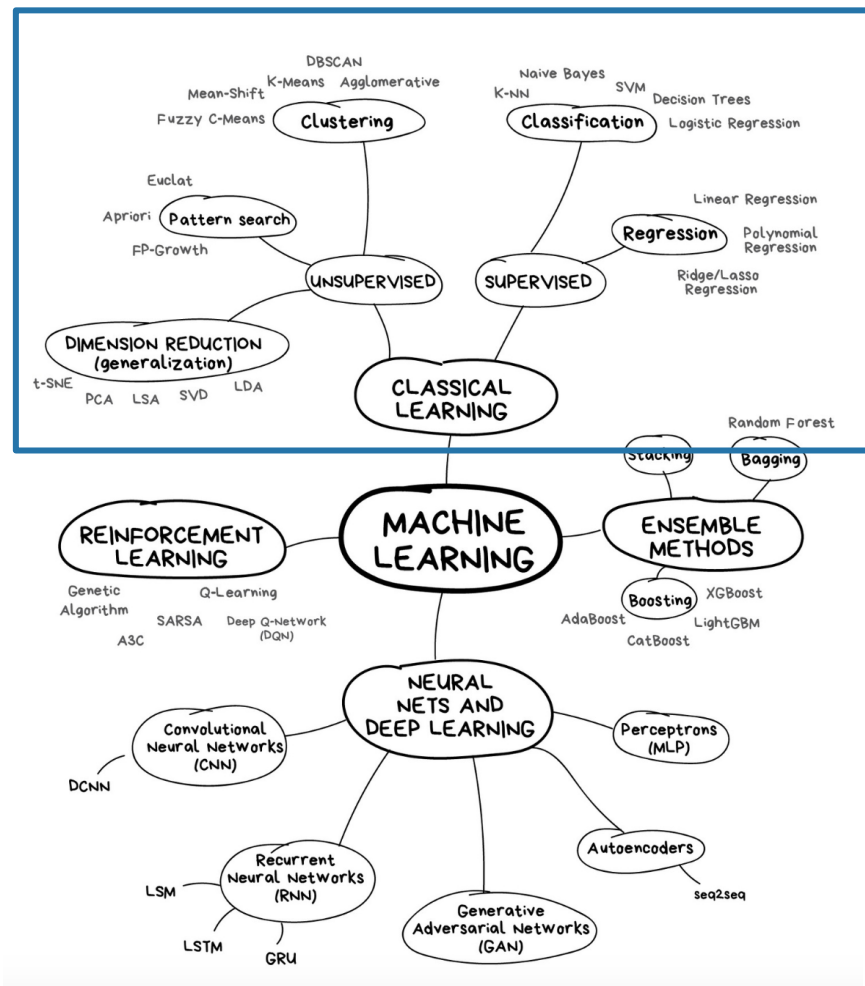
All subsets of n columns
is 2^n models!

Pick the model that performs best on
validation data...

Now retrain that column subset on a
combo of training set + validation set

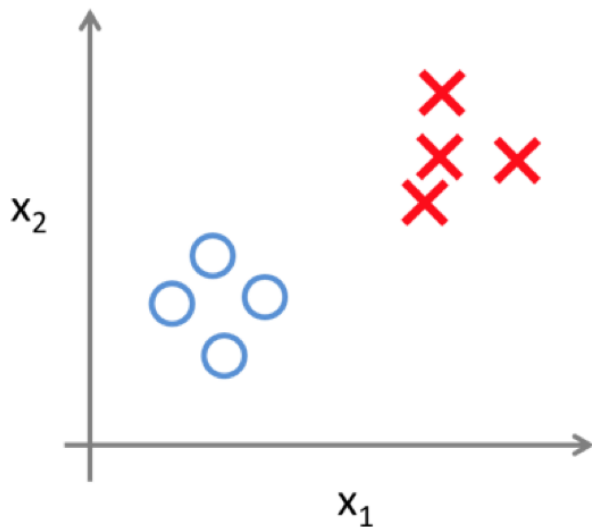
And evaluate its future performance
level on the test set





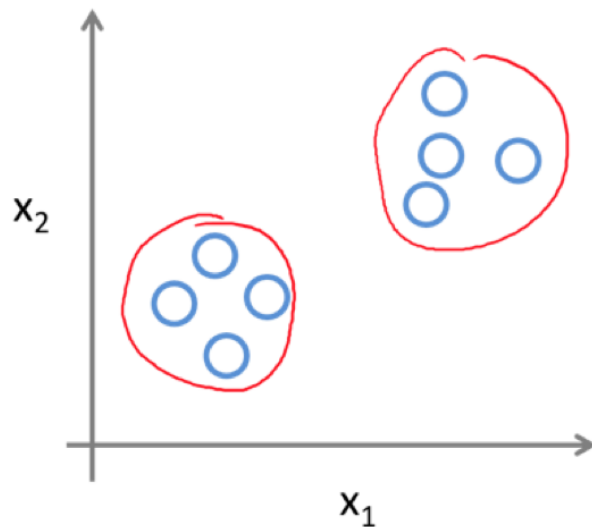
Two modes of machine learning

Supervised Learning



You give examples, the computer learns from them

Unsupervised Learning



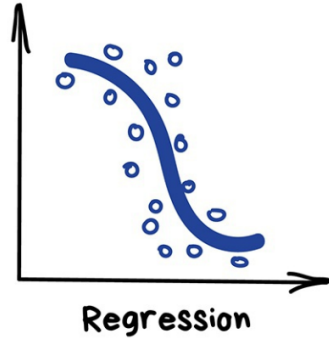
The computer determines how to classify based on properties within the data

Regression

"Draw a line through these dots. Yep, that's the machine learning"

Today this is used for:

- Stock price forecasts
- Demand and sales volume analysis
- Medical diagnosis
- Any number-time correlations



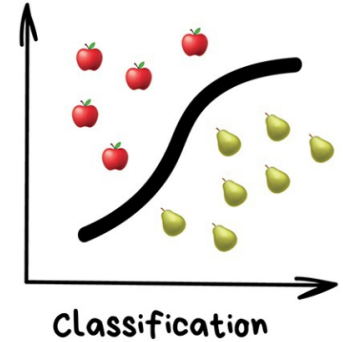
continuous variable prediction

Classification

"Splits objects based at one of the attributes known beforehand. Separate socks by based on color, documents based on language, music by genre"

Today used for:

- Spam filtering
- Language detection
- A search of similar documents
- Sentiment analysis
- Recognition of handwritten characters and numbers
- Fraud detection

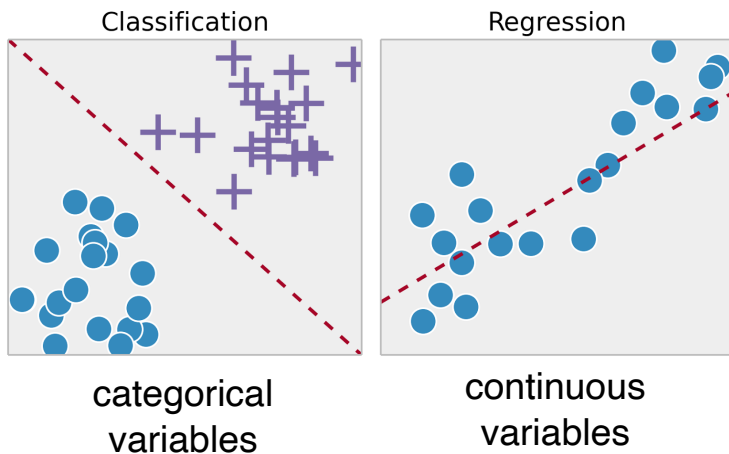


Popular algorithms: Naive Bayes, Decision Tree, Logistic Regression, K-Nearest Neighbours, Support Vector Machine

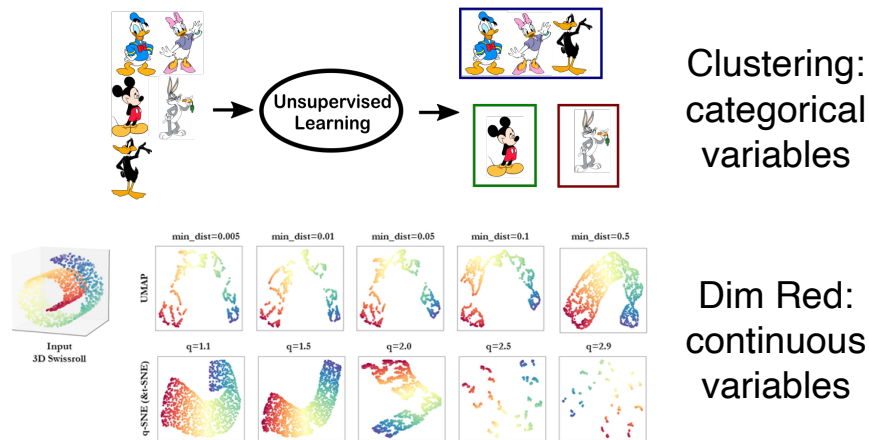
categorical variable prediction

Approaches to machine learning

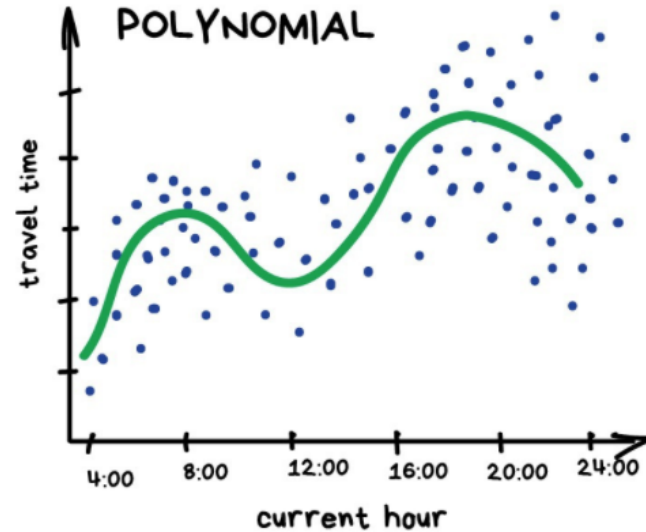
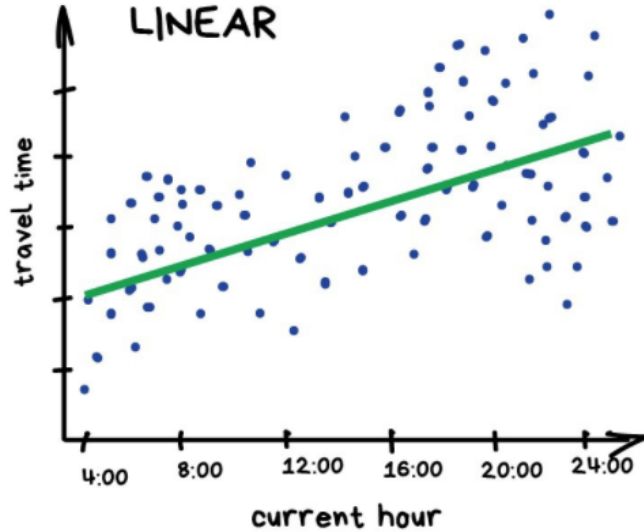
Supervised Learning



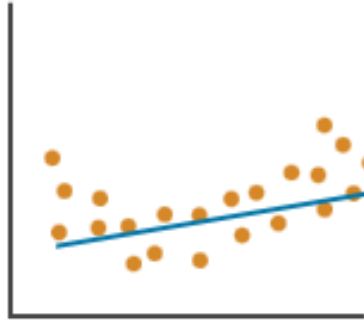
Unsupervised Learning



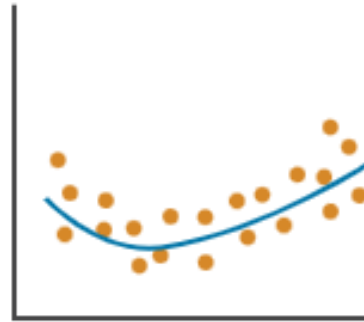
PREDICT TRAFFIC JAMS



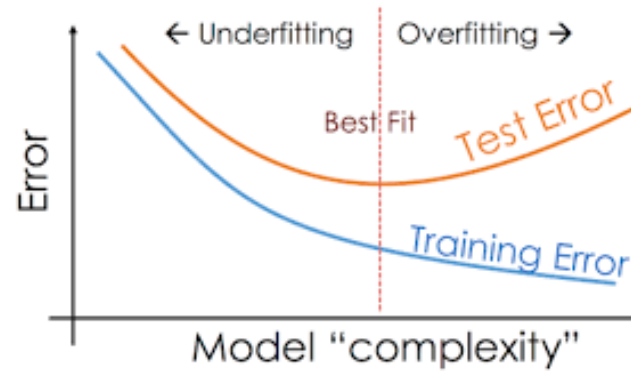
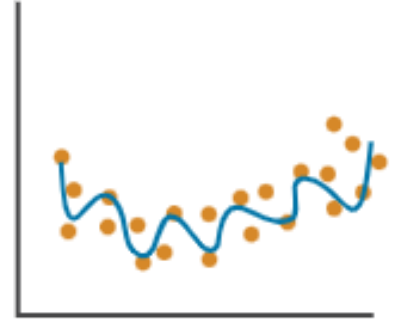
Underfit



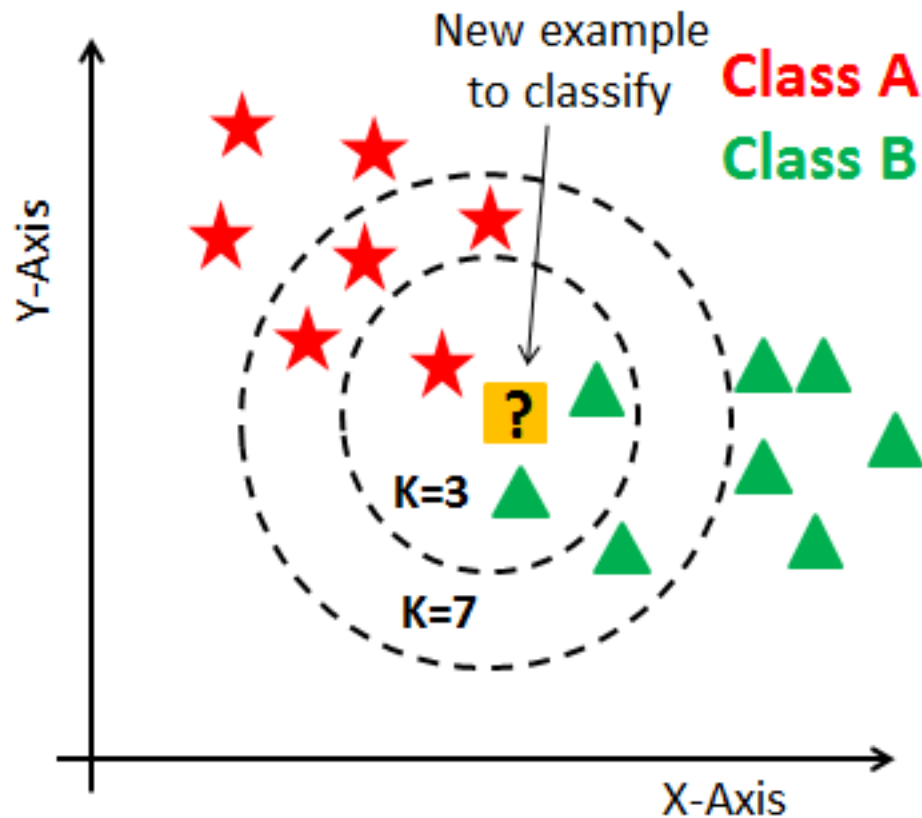
Best fit?



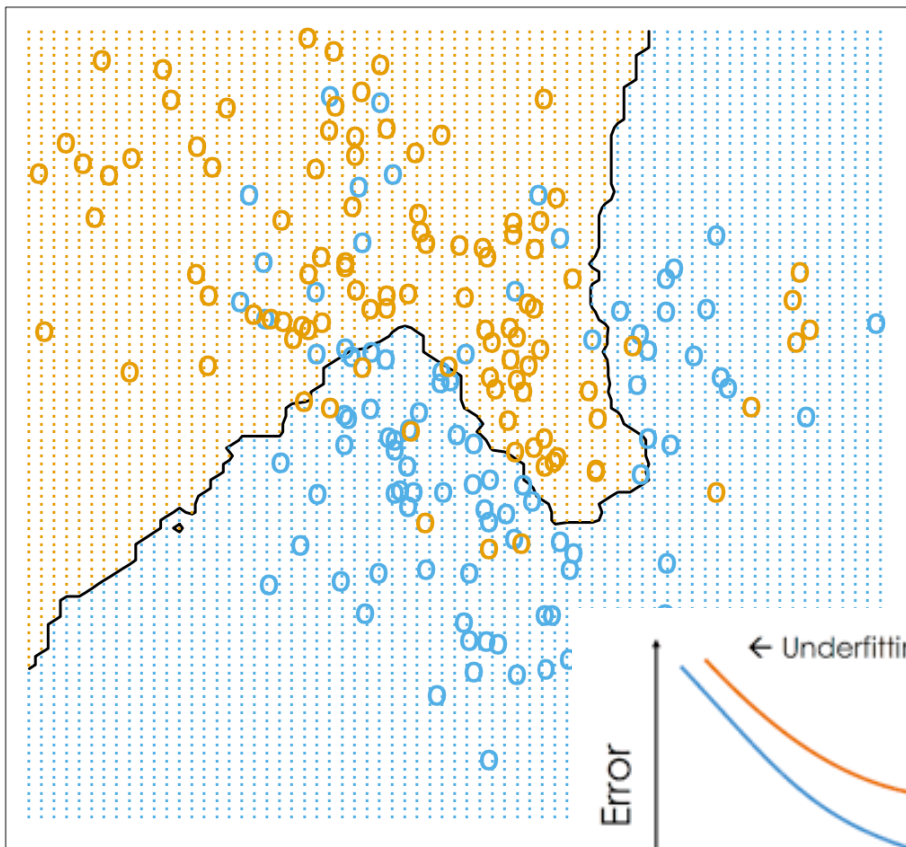
Overfit



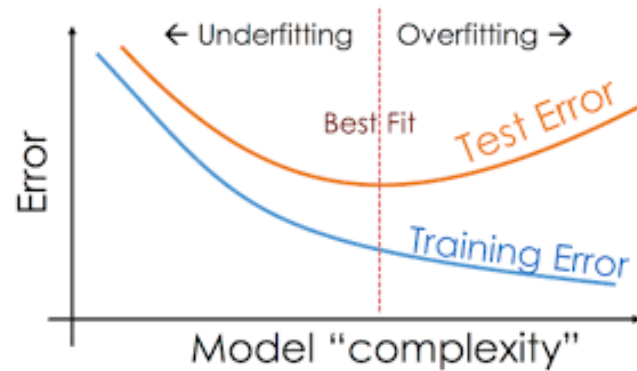
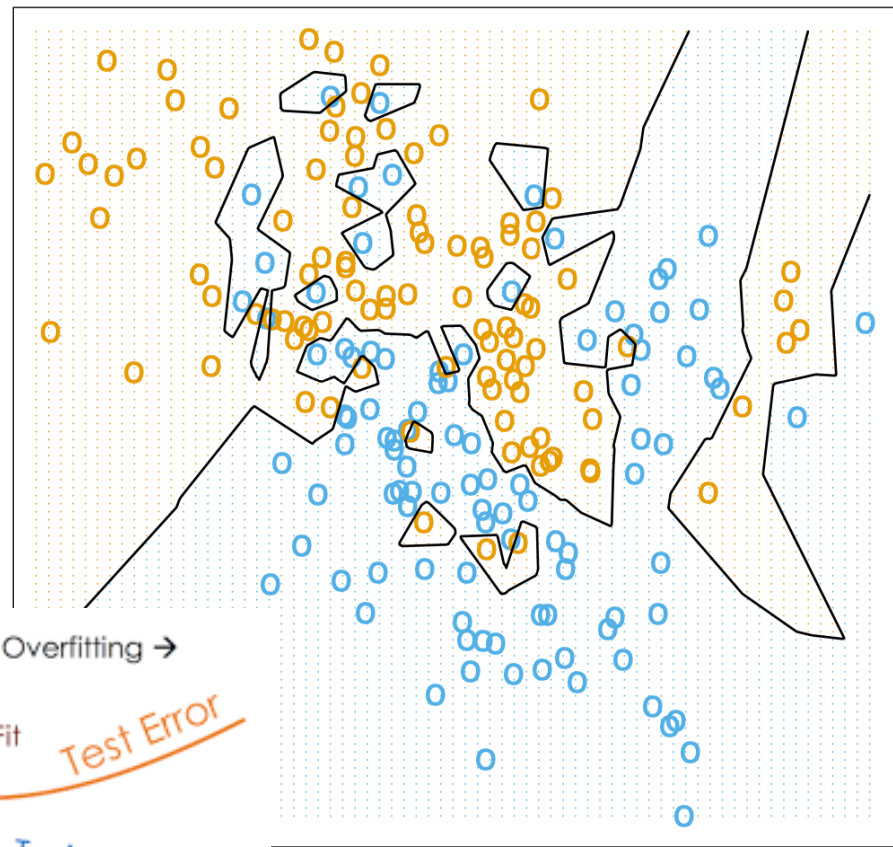
K-nearest neighbors



15-Nearest Neighbor Classifier

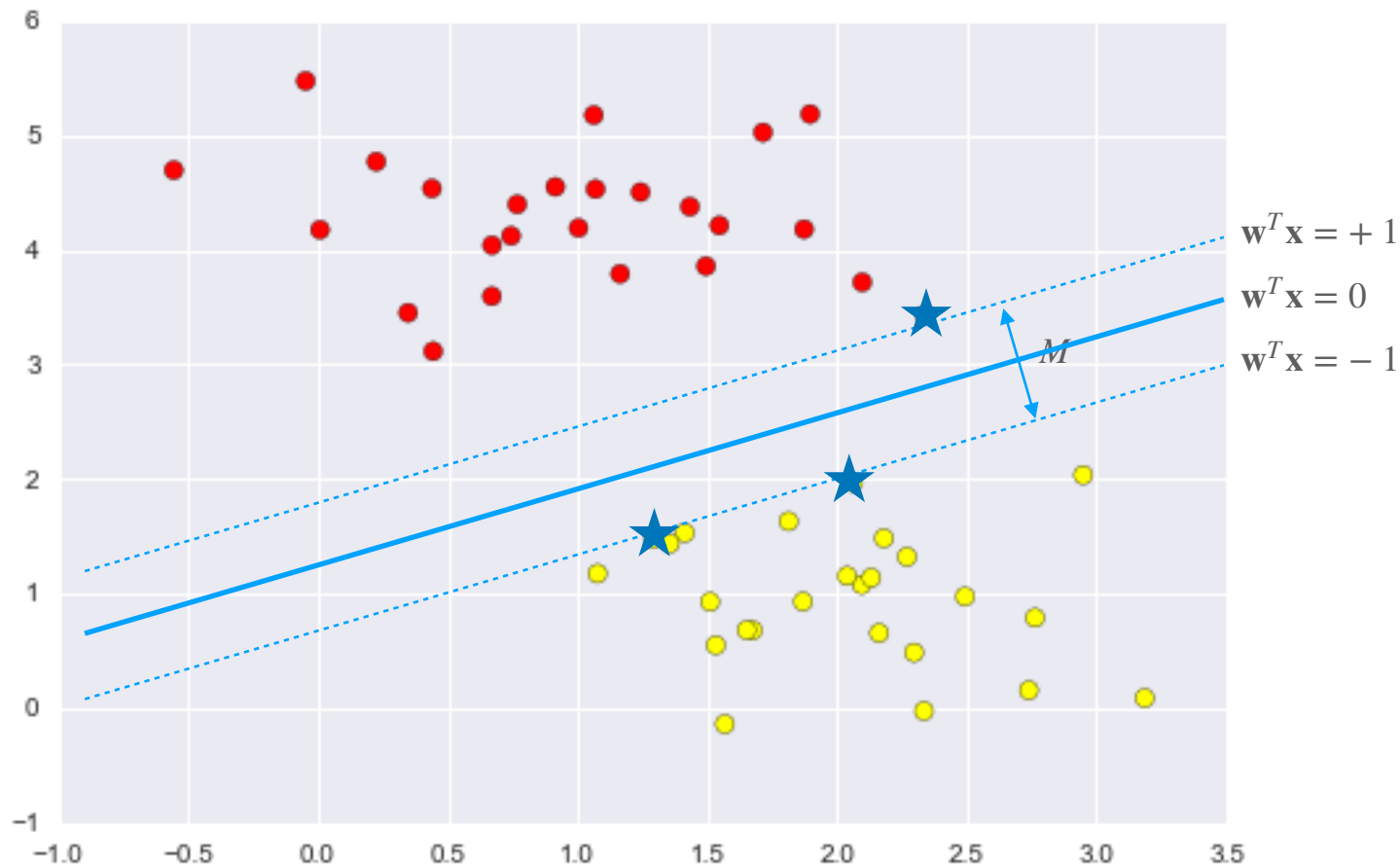


1-Nearest Neighbor Classifier



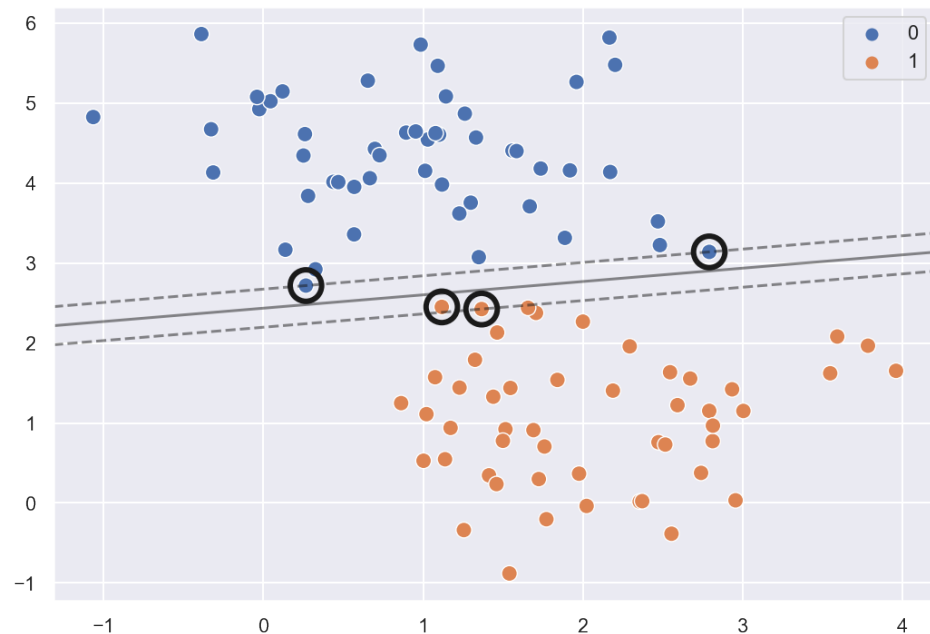
Support Vector Machines:

Find the support
data points that
maximize the
margin M



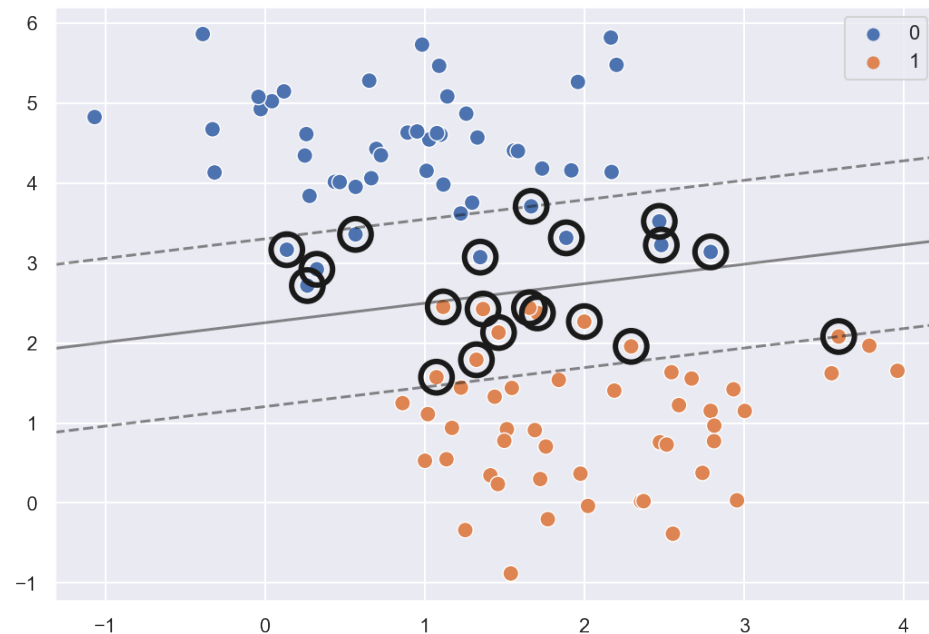
Hard margin

$C = 10.0$



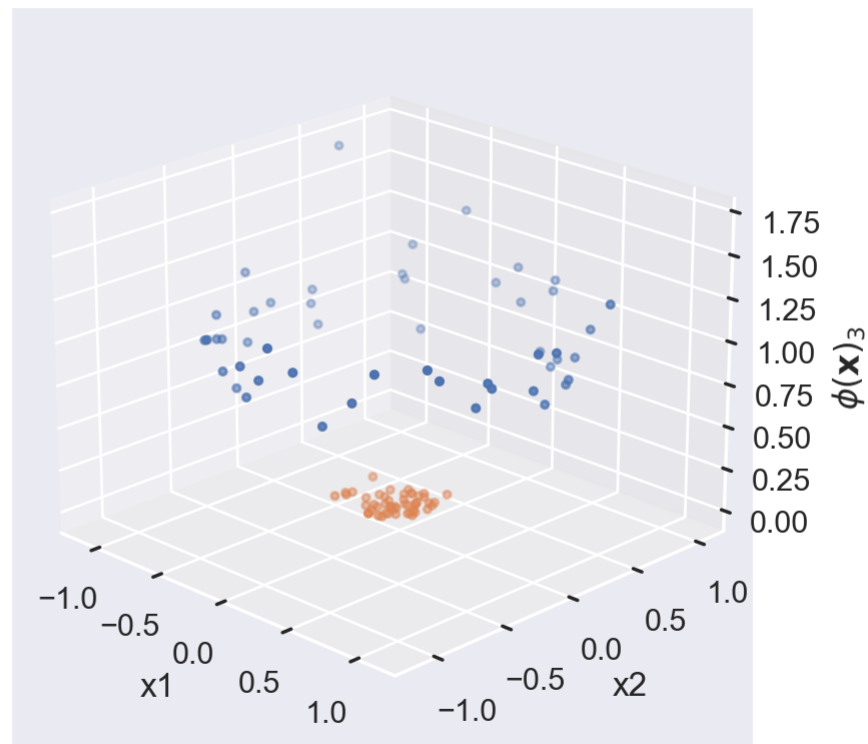
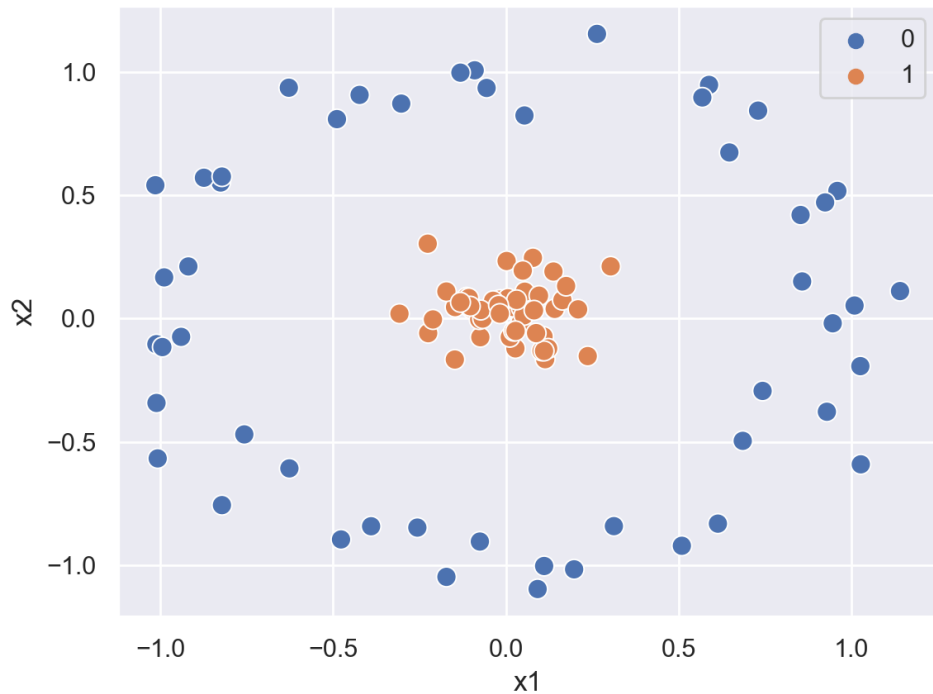
Soft margin

$C = 0.1$



Find: $\arg \min_{\mathbf{w}} \frac{1}{2} \|\mathbf{w}\|^2 + C \times \sum_{i=1}^n (1 - y_i \times (\mathbf{w}^T \mathbf{x}_i + b))_+$

$$\phi(\mathbf{x}) = [x_1, x_2, x_1 \cdot x_2 + x_1^2 \cdot x_2^2]$$



More generally this is the polynomial kernel

